

Course Code: (CSE225)	Course Title: Introduction to Combinatorics and Graph Theory Type of Course: Program Core - Theory	L- T- P- C	3 - 0	0	3
Version No.	version 1				
Course Pre-requisites	Basic logic and Set theory				
Anti-requisites	nil				
Course Description	Graph Theory is a blend of the mathematical techniques applicable to Computer science, Information Technology and Statistics. Graph Theory gives us, both an easy way to pictorially represent many major mathematical results, and insights into the deep theories behind them. In this course, among other intriguing applications, we will see how GPS systems find shortest routes, how engineers design integrated circuits, how biologists assemble genomes, why a political map can always be colored using a few colors. Topics Include: Principles of Inclusion and Exclusion, Rook Polynomial, Derangements. Graph Theory: Graph Terminologies, Isomorphism, Coloring, Matching, Planar Graphs, Trees Terminologies, Traversals, Spanning Trees, Shortest path algorithms, Prefix Codes				
Course Objective	The objective of the course is to familiarize the learners with the concepts : Introduction to Combinatorics and Graph Theory and attain Skill Development through Participative Learning techniques.				
Course Outcomes	CO1: Explain the fundamental concepts of Graph theory. [L1: Knowledge] CO2: Discuss theorems of matching, connectivity, coloring and planar graphs. [L2: Comprehension] CO3: Discuss different types of trees and traversal techniques. [L2: Comprehension] CO4: Apply different algorithms to find optimal path for a given graph. [L3: Applications]				
Course Content:					
Module 1	Introduction to Graph Theory	Assignment	Data Collection	07 Sessions	

Introduction to Graph Theory 07H [Knowledge Level]
 Basic Concepts: definition, types of graphs, Graph Terminology and Special Types of Graph, representation of a graph and connectedness graph: (paths, walk. cycles, edge deleted and vertex deleted).

Module 2	Introduction to Graph Theory contd	Assignment	Analysis of test results and also can be dealt with Lab	11 Sessions
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Introduction to Graph Theory contd. 11H [Comprehension Level]
 Graph isomorphism, Eulerian graph, Hamiltonian graph, Planar graph (three utility problem), Graph coloring, Combinatorics-Principle of Inclusion and Exclusion.

Module 3	Trees	Assignment	MS Excel, Using Graphs and Pi Charts and tables for analysis	13 Sessions
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Trees 13H [Comprehension Level] Tree: Definitions, properties, Rooted trees, Binary search tree, Decision tree, prefix code, Tree traversal: in-order, pre-order, post-order, infix, postfix, prefix, spanning tree: BFS, DFS.

Module 3	Algorithm on networks	Assignment	MS Excel, Using Graphs and Pi Charts and tables for analysis	13 Sessions	Assignment	MS Excel, Using Graphs and Pi Charts and tables for analysis	13 Sessions
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Algorithm on networks Shortest path algorithm- Dijkstra's algorithm, Minimal spanning tree- Kruskal algorithm and Prim's algorithm, Transport network- Max-flow/Min-cut algorithm, Combinatorics-Rook polynomial, Derrangements .